SURYA: QUICK REFERENCE GUIDE

Functions, Purposes & Applications

1: WHAT IS SURYA?

Surya = Sanskrit for "Sun"

Definition:
First foundation model for heliophysics and space weather prediction

Key Stats:

- 🔹 366 million parameters
- 🔹 Trained on 9 years of SDO data (2011-2019)
- 🔹 Processes 257 terabytes of solar observations
- 🔹 4096×4096 pixel resolution
- 🔹 13 channels (8 AIA + 5 HMI)

Developers:
IBM Research + NASA + 8 research institutions

Released: August 2024 (Open Source)

2: WHY SURYA MATTERS

The Problem:

☀️ Solar storms can:

- Endanger astronauts
- Damage satellites ($100M+ losses)
- Knock out power grids (billions in damage)
- Disrupt GPS and communications
- Ground polar flights

Current Limitation:

⏰ **Only 1-hour warning** for solar flares

Surya's Solution:

✅ **2-hour advance warning** (100% improvement!)

3: THE 4 MAIN FUNCTIONS

1️⃣ SOLAR FLARE FORECASTING

What: Predict M/X-class flares 24 hours ahead
Performance: TSS = 0.436 (16% better than baseline)
Lead Time: 2 hours (vs 1 hour traditional)
Use: Protect astronauts, satellites, power grids

2️⃣ ACTIVE REGION SEGMENTATION

What: Auto-identify solar active regions
Performance: IoU = 0.768 (12% better than UNet)
Use: Track flare sources, predict emergence

3️⃣ EUV SPECTRA PREDICTION

What: Predict 1343-dimensional EUV irradiance
Performance: MAPE = 1.48%
Use: Satellite drag modeling, orbital predictions

4️⃣ SOLAR WIND FORECASTING

What: Predict solar wind speed 4 days ahead
Performance: RMSE = 75.92 km/s
Use: Geomagnetic storm warnings, power grid prep

4: PRIMARY PURPOSES

Purpose 1: SPACE WEATHER FORECASTING

🎯 Enable protective actions before solar storms hit
🎯 2-hour warning for critical infrastructure
🎯 4-day ahead solar wind predictions

Purpose 2: SCIENTIFIC DISCOVERY

🔬 Automate analysis of 15 years of SDO data
🔬 Extract patterns from petabytes automatically
🔬 Accelerate heliophysics research

Purpose 3: FOUNDATION MODEL FOR SPACE PHYSICS

🚀 First demonstration of foundation models in space science
🚀 Single model → multiple tasks (transfer learning)
🚀 Methodology for other space applications

SLIDE 5: REAL-WORLD APPLICATIONS

🛰️ SPACE MISSIONS

Application: Astronaut & spacecraft protection
Action: 2-hour warning → cancel EVAs, enter safe mode
Impact: Reduced radiation exposure, equipment protection

⚡ POWER GRIDS

Application: Geomagnetic storm preparation
Action: 4-day warning → reduce transformer load, stage crews
Impact: Prevent blackouts (Quebec 1989: $2B damage)

🛰️ SATELLITES

Application: Orbital prediction & collision avoidance
Action: EUV forecast → atmospheric density → adjust orbits
Impact: Saved SpaceX $100M in 2022 (could have)

✈️ AVIATION

Application: Polar route management
Action: Flare warning → reroute flights
Impact: Passenger safety, communication reliability

📡 GPS/NAVIGATION

Application: Positioning accuracy maintenance
Action: Predict ionospheric disturbances
Impact: Military, autonomous vehicles, emergency services

6: TECHNICAL INNOVATION

Architecture:

✅ Vision Transformer with long-short attention
✅ Spectral gating (5% memory reduction)
✅ Handles 4096×4096 resolution (10× typical models)

Training:

✅ Two-phase: One-step → autoregressive
✅ 160,000 steps on 128 GPUs
✅ Self-supervised learning (no manual labels)

Data:

✅ 13 SDO channels (AIA + HMI)
✅ 12-minute cadence, 9 years
✅ First ML-ready multi-instrument solar dataset

7: PERFORMANCE COMPARISON

Task	Surya	Previous Best	Improvement
Solar Flare	TSS: 0.436	TSS: 0.358	**+16%** ⬆️
AR Segmentation	IoU: 0.768	IoU: 0.688	**+12%** ⬆️

Task	Surya	Previous Best	Improvement
EUV Spectra	MAPE: 1.48%	N/A	**Best** ✅
Solar Wind	RMSE: 75.92	N/A	**4-day lead** ✅

Key Achievement: 2-hour flare warning (vs 1-hour traditional)

8: vs TRADITIONAL APPROACHES

Traditional Physics Models:

❌ Years to develop
❌ Hours per forecast
❌ Requires code rewrite for new tasks
✅ Interpretable (physics equations)

Surya Foundation Model:

✅ Weeks to train
✅ Minutes per forecast
✅ Fine-tune for new tasks (hours)
❌ Less interpretable (learned features)

Traditional Task-Specific ML:

❌ Separate model per task
❌ 10,000+ training samples needed
❌ Days-weeks training per task

Surya Foundation Model:

✅ Single model for all tasks
✅ 1,000+ samples (fine-tuning)
✅ Hours to adapt to new task

9: CASE STUDIES

Case 1: SpaceX Starlink Loss (Feb 2022)

Event: 40 satellites lost ($100M) due to geomagnetic storm
Surya Could Have:
✅ Predicted atmospheric expansion 4 days ahead
✅ Delayed launch by 48 hours
✅ Saved entire satellite batch

Case 2: Quebec Blackout (1989)

Event: 9-hour blackout, 6 million affected, $2B damage
Surya Could Have:
✅ 4-day warning of solar wind arrival
✅ Pre-positioned repair crews
✅ Reduced transformer loading
✅ Prevented or minimized blackout

Case 3: Aviation Rerouting

Event: X-class flare disrupts polar communications
Surya Does:
✅ 2-hour warning to airlines
✅ Reroute 20+ flights to lower latitudes
✅ Maintain passenger safety

10: IMPACT & SIGNIFICANCE

Economic:

💰 Prevents $1-2 trillion in space weather damages
💰 Protects critical infrastructure
💰 Enables safer space operations

Scientific:

🔬 First foundation model for space physics
🔬 Automates petabyte-scale data analysis
🔬 Opens doors for AI in other sciences

Societal:

👩‍🚀 Protects astronauts & space explorers
🌍 Maintains GPS, power, communications
🚀 Enables Mars missions & deep space travel

Technological:

🤖 Demonstrates AI for scientific discovery
🤖 Transfer learning in space domain
🤖 Foundation for future space AI

11: ACCESSIBILITY & DEPLOYMENT

Open Source:

✅ GitHub: https://github.com/NASA-IMPACT/Surya
✅ HuggingFace: nasa-ibm-ai4science/Surya-1.0
✅ Apache 2.0 License

Integration Options:

1. Direct use via HuggingFace Transformers
2. IBM TerraTorch library
3. SuryaBench framework

Requirements:

- **Inference:** NVIDIA V100/A100 GPU
- **Fine-tuning:** 4× A100 GPUs, 2-24 hours
- **Full training:** 128× A100 GPUs (not needed)

12: FUTURE DIRECTIONS

Near-Term (1-2 years):

- Multi-mission data integration (STEREO)
- Extended prediction horizons (48-hr flares, 7-day wind)

- Uncertainty quantification
- Real-time NOAA deployment

Long-Term (3-5 years):

- Physics-informed hybrid models
- CME arrival time prediction
- Application to other stars
- On-board spacecraft AI

Ultimate Vision:

🌟 Standard tool for all space weather operations
🌟 Enable safe deep space exploration
🌟 Protect Earth's space infrastructure

13: KEY QUOTES

> *"Pulling features and events out of petabytes of data is a laborious process and now we can automate it."*
> — **Dr. Kathy Reeves**, Harvard-Smithsonian Center for Astrophysics

> *"Our hope is that the model has learned all the critical processes behind our star's evolution through time so that we can extract actionable insights."*
> — **Dr. Andrés Muñoz-Jaramillo**, SouthWest Research Institute

> *"With Surya we have created the first foundation model to look the Sun in the eye and forecast its moods."*
> — **Juan Bernabé-Moreno**, IBM Research Director

14: SUMMARY - THE BIG PICTURE

What Surya Does:

✅ Predicts solar activity with unprecedented lead time
✅ Automates analysis of 15 years of solar data
✅ Enables protective actions before storms hit

How It Works:

✅ 366M parameter vision transformer
✅ Trained on full-resolution SDO data
✅ Transfer learning for multiple tasks

Why It Matters:

✅ Protects $1-2 trillion in infrastructure
✅ Enables safer space exploration
✅ Advances scientific discovery

Impact:

✅ 2-hour flare warning (vs 1-hour)
✅ 4-day solar wind forecasts
✅ First foundation model for space physics

15: QUICK FACTS

Model Size: 366 million parameters
Training Data: 9 years (2011-2019)
Data Volume: 257 terabytes
Resolution: 4096×4096 pixels
Channels: 13 (8 AIA + 5 HMI)
Training Hardware: 128 GPUs
Training Duration: ~2 weeks
Inference Speed: 2 seconds/prediction
Developers: IBM + NASA + 8 institutions
License: Open source (Apache 2.0)
Release Date: August 2024

Performance Highlights:

- TSS: 0.436 (solar flare)
- IoU: 0.768 (segmentation)
- MAPE: 1.48% (EUV)
- RMSE: 75.92 km/s (solar wind)

16: REFERENCES

Main Sources:

1. IBM Research Blog:
   https://research.ibm.com/blog/surya-heliophysics-ai-model-sun

2. GitHub Repository:
   https://github.com/NASA-IMPACT/Surya

3. Model Weights:
   https://huggingface.co/nasa-ibm-ai4science

4. SuryaBench:
   https://huggingface.co/collections/nasa-impact/suryabench

END OF PRESENTATION GUIDE